

Department of Electronics and Communication Engineering
B. Tech. – ECE & VDT

Semester: III**Max. Marks: 150**

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Marks are indicative of word limit.

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Q 5	In the context of a basic computer, with the help of a timing diagram explain the working of the microoperation: $C_7T_3: SC \leftarrow 0$.	10
Q 6	The content of AC in the basic computer is hexadecimal A937 and the initial value of E is 1. Determine the contents of AC, E, PC, AR, and IR in hexadecimal after the execution of the CLA instruction.	10
Q 7	The content of PC in the basic computer is 3AF. The content of AC is 7EC3. The content of memory at address 3AF is 932E. The content of memory at address 32E is 09AC. The content of memory at address 9AC is 8B9F. a. What is the instruction that will be fetched and executed next? b. Show the binary operation that will be performed in the AC when the instruction is executed.	10
Q 8	A computer uses a memory of 65,536 words with eight bits in each word. It has the following registers: PC, AR, TR (16 bits each), and AC, DR, IR (eight bits each). A memory-reference instruction consists of three words: an 8-bit operation-code (one word) and a 16-bit address (in the next two words). All operands are eight bits. There is no indirect bit. List the sequence of microoperations for fetching a memory reference instruction and then placing the operand in DR. Start from timing signal T0.	10
Q 9	The operations to be performed with a flip-flop F (not used in the basic computer) are specified by the following register transfer statements: $xT_3: F \leftarrow 1$ Set F to 1 $yT_1: F \leftarrow 0$ Clear F to 0 $zT_2: F \leftarrow F'$ Complement F $wT_5: F \leftarrow G$ Transfer value of G to F Otherwise, the content of F must not change. Draw the logic diagram showing the connections of the gates that form the control functions and the inputs of flip flop F. Use a JK flip-flop and minimize the number of gates.	10
Q 10	Derive the control gates associated with the program counter PC in the basic computer. Draw the logic diagram of the gates and show how the output is connected to the LD, INR and CLR inputs of PC. Minimize the number of gates.	10

Q 11	In certain scientific computations it is necessary to perform the arithmetic operation $(A_i + B_i) (C_i + D_i)$ with a stream of numbers. Specify a pipeline configuration to carry out this task. List the contents of all registers in the pipeline for $i = 1$ through 6.	10
Q 12	Determine the number of clock cycles that it takes to process 200 tasks in a six-segment pipeline.	10
Q 13	A three segment pipeline structure is used to perform the arithmetic operation $(A_i * B_i) + C_i$ for $i = 1$ through 6 with a stream of numbers. R1 through R5 are registers that receive new data with every clock pulse. The pipeline configuration has the following propagation times: 40 ns for the operands to be read from memory into registers R1 and R2, 45 ns for the signal to propagate through the multiplier, 5 ns for the transfer into R3, and 15 ns to add the two numbers into R5. a. What is the minimum clock cycle time that can be used? b. A non-pipeline system can perform the same operation by removing R3 and R4. How long will it take to multiply and add the operands without using the pipeline? c. Calculate the speedup of the pipeline for 10 tasks and again for 100 tasks. d. What is the maximum speedup that can be achieved?	10
Q 14	Explain four possible hardware schemes that can be used in an instruction pipeline in order to minimize the performance degradation caused by instruction branching.	10
Q 15	Consider the four instructions in the following program. Suppose that the first instruction starts from step 1 in the four-segment pipeline. Specify what operations are performed in the four segments during step 4. Load $R1 \leftarrow M[312]$ ADD $R2 \leftarrow R2 + M[313]$ INC $R3 \leftarrow R3 + 1$ STORE $M[314] \leftarrow R3$	10